

KHANG THAI

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EDUCATION

University of Southern California	Los Angeles, CA
Master of Science in Applied Data Science	August 2025-May 2027
• Relevant Coursework: Machine Learning, Data Management, Deep Learning	
University of California, Los Angeles	Los Angeles, CA
Bachelors of Science in Statistics and Data Science	September 2023-June 2025
• Relevant Coursework: Statistical Programming with R, Python for Data Science, Statistical Models and Data Mining, Data Analysis and Regression, Computation and Optimization for Statistics, Monte Carlo Methods, Statistical Consulting, Databases	
San Diego Miramar College	San Diego, CA
Associate of Science	August 2021-May 2023
• Dean's List 2021 - 2023	

EXPERIENCE

Hologic, Inc.	San Diego, CA
Bioinformatics Intern	June 2026-Present
• Architected a hybrid Retrieval-Augmented Generation (RAG) pipeline combining dense vector search (Qdrant), lexical matching (BM25), and cross-encoder neural reranking (bge-reranker) over 16,000+ controlled assay document chunks.	
• Built an agentic orchestration layer via LangGraph to manage stateful multi-step reasoning, dynamic tool selection, and automated fallback routing across complex document analysis tasks.	
• Implemented human-in-the-loop (HITL) approval gates and structured Pydantic schema validation to mitigate hallucinations and meet rigorous bioinformatics regulatory standards.	
• Designed scalable document ingestion, indexing microservices, streamlining document processing workflows and accelerating knowledge discovery across internal assay datasets.	
NeverEnding, Inc	Remote
Software Development Backend Intern	May 2024-August 2024
• Engineered secure backend payment systems in Python, integrating Stripe API to manage subscription and transaction workflows	
• Designed and deployed scalable database schemas using Django and JSON to manage user accounts and digital gifting services	
• Collaborated across teams using Agile methodologies in Jira to build, test, and ship responsive web features via GitHub	

PROJECTS

Diabetes Risk Predictive Modeling	March 2025
• Constructed a predictive pipeline utilizing classification algorithms on 9,500+ clinical records, achieving a 97% accuracy rate	
• Evaluated model stability using 5-fold cross-validation and feature importance analysis to ensure generalization and interpretability	
• Visualized clinical risk factors and key feature interactions to extract actionable epidemiological insights	
LA Airbnb Analysis	May 2024
• Applied Natural Language Processing (NLP) and Latent Dirichlet Allocation (LDA) to extract latent themes and pain points from unstructured guest reviews	
• Developed predictive models (logistic regression and random forest) to identify key structural drivers of Superhost status	

TECHNOLOGIES

AI & Machine Learning: Large Language Models (LLMs), Hybrid RAG (Dense + BM25), Cross-Encoder/Neural Reranking, Agentic Workflows & Multi-Agent Orchestration, Structured Schema Validation, Prompt Engineering, NLP, Supervised/Unsupervised Learning, Statistical Modeling

Frameworks & Libraries: LangGraph, LangChain, PyTorch, Scikit-Learn, Pydantic, FastAPI, Django

Databases & Vector Search: Qdrant, PostgreSQL, MySQL

Languages, Tools & Platforms: Python, R, SQL, Java, JavaScript, Git, Jira, Linux/Bash, Tableau, Docker